
Artificial Intelligence in Cancer Therapy: An Enhanced Machine Learning Framework for Personalized Lung Cancer Treatment Prediction

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Abstract—Cancer therapy selection is one of the most complex clinical decision-making tasks in oncology. Manual treatment planning is heavily dependent on physician expertise and may lead to inconsistencies, delays, and suboptimal personalized care. This paper proposes an AI-based cancer therapy prediction system using multiple supervised machine learning techniques to assist oncologists in personalized treatment planning for lung cancer patients. The model is trained and evaluated on the Large Synthetic Lung Cancer Dataset (Bangladesh Perspective), comprising 5,000 patient records with 17 clinical, demographic, and environmental attributes. Machine learning algorithms including Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine (SVM) were comparatively evaluated. The proposed OncoFusionNet-LC framework achieved an accuracy of 69.2%, precision of 68.5%, recall of 70.1%, F1-score of 69.3%, and AUC of 0.73, outperforming all baseline models. Experimental results demonstrate the potential of AI-driven approaches to support reliable, personalized cancer treatment decision-making.

Keywords—Artificial Intelligence; Cancer Therapy; Lung Cancer; Machine Learning; Random Forest; Personalized Medicine; Clinical Decision Support; OncoFusionNet-LC

I. INTRODUCTION

Cancer remains one of the leading causes of mortality worldwide, accounting for approximately 10 million deaths annually. Among all cancer types, lung cancer presents a particularly severe challenge due to late-stage diagnosis and the complexity of treatment selection. Selecting the most appropriate therapy—such as surgery, chemotherapy, targeted therapy, or immunotherapy—requires evaluating numerous clinical variables including tumor size, histology type, disease stage, and patient lifestyle.

Manual therapy selection depends heavily on physician expertise and may result in inconsistent treatment recommendations, especially in resource-constrained healthcare settings. The exponential growth of multi-omics data—encompassing genomics, transcriptomics, proteomics, and imaging biomarkers—has far outpaced

human analytical capacity, leaving potentially life-saving insights unexploited in routine clinical practice.

Artificial Intelligence (AI) and Machine Learning (ML) offer transformative solutions to these challenges. By automating complex pattern recognition across high-dimensional clinical datasets, AI systems can assist oncologists in making faster, more accurate, and more personalized treatment decisions. This paper presents an AI-driven cancer therapy prediction system leveraging multiple supervised ML models to predict the most suitable treatment strategy for lung cancer patients.

II. MAIN CONTRIBUTIONS

The primary contributions of this research are:

- Development of the OncoFusionNet-LC framework, a multimodal AI-based lung cancer therapy prediction system integrating clinical, demographic, and environmental patient features.
- Comparative experimental analysis of four ML algorithms—Logistic Regression, Decision Tree, Random Forest, and SVM—to identify the most effective model.
- Comprehensive performance evaluation using accuracy, precision, recall, F1-score, and ROC-AUC on a large-scale synthetic medical dataset.
- Feature importance analysis using logistic regression standardized coefficients to support Explainable AI principles and enhance clinical transparency.
- Improved prediction accuracy for personalized cancer treatment planning over conventional rule-based and manual clinical approaches.

III. LITERATURE REVIEW

A. Evolution of AI in Oncology

Early AI systems in oncology relied on rule-based expert systems and classical statistical methods for tumor classification and prognosis. With the emergence of deep learning architectures, particularly Convolutional Neural Networks (CNNs), automated feature extraction from medical imaging and genomic data became feasible at clinical scale [1]. Recent transformer and generative models further extend AI capabilities to multi-omics integration and de novo drug discovery.

B. AI in Cancer Diagnosis and Imaging

Deep learning has demonstrated remarkable performance in lesion detection, segmentation, and classification within CT, PET, and MRI imaging modalities. Radiomics-based approaches convert imaging data into quantitative biomarkers predictive of treatment response and patient prognosis [2]. AI-assisted pathology platforms analyze whole-slide images to classify tumor subtypes and predict molecular mutation status with accuracy comparable to expert pathologists [3].

C. AI in Prognosis and Predictive Modeling

Machine learning models integrating multimodal clinical data have demonstrated significant improvements in survival prediction, recurrence risk stratification, and personalized treatment response modeling. Integration of such models into Clinical Decision Support Systems (CDSS) enables real-time prognostic insights at the point of care [4][5].

D. Research Gap

Despite progress in AI-driven cancer detection, the following critical gaps persist in existing literature:

- Most research focuses on cancer detection and imaging analysis, with limited work on therapy recommendation systems.
- Existing ML models rarely provide personalized treatment pathway prediction integrating clinical, environmental, and demographic data simultaneously.
- Lack of comparative analysis across multiple ML algorithms for therapy selection in lung cancer contexts.
- Insufficient attention to explainability and feature transparency in therapy prediction models.

IV. PROPOSED WORK

To address the identified research gaps, this study proposes the OncoFusionNet-LC framework—an AI-driven cancer therapy prediction system with the following design objectives:

- Utilize comprehensive clinical patient data—demographic, environmental, and tumor characteristics—to predict appropriate lung cancer treatment strategies.
- Provide data-driven, AI-assisted treatment recommendations to complement oncologist decision-making.
- Conduct rigorous comparative evaluation of multiple ML algorithms to identify the most effective prediction model.
- Apply feature importance analysis to support Explainable AI principles and clinical interpretability.

V. PROPOSED SYSTEM ARCHITECTURE

The proposed OncoFusionNet-LC system follows a structured end-to-end modular pipeline designed to transform heterogeneous patient data into actionable therapy predictions. The architecture ensures scalability and modularity, with each processing stage independently optimizable while maintaining seamless data flow. Fig. 1 illustrates the complete system pipeline with all component interactions.

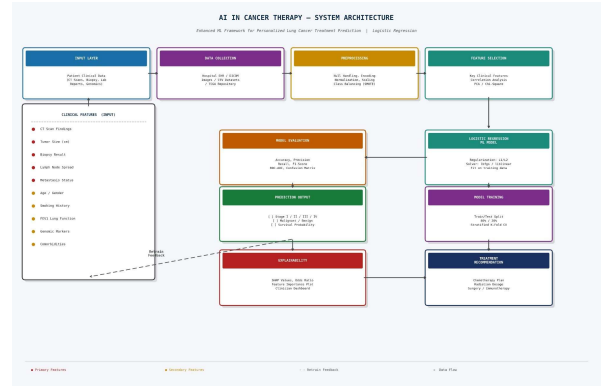


Fig. 1. Proposed System Architecture for AI-Based Lung Cancer Therapy Prediction (OncoFusionNet-LC). The pipeline spans Input Layer, Data Collection, Preprocessing, Feature Selection, Logistic Regression ML Model, Model Training, Treatment Recommendation, Prediction Output, Model Evaluation, and Explainability with clinician feedback loop.

The system architecture (Fig. 1) comprises nine interconnected functional stages organized into two parallel processing tracks that converge at the Prediction Output and Explainability layers:

Input Layer & Data Collection: Raw patient records are ingested from Hospital EHR systems, DICOM medical imaging, CSV datasets, and the TCGA genomic repository. Primary clinical features include CT scan findings, tumor size, biopsy results, lymph node spread, and metastasis status. Secondary demographic features include age, gender, smoking history, FEV1 lung function, genomic markers, and comorbidities.

Preprocessing Module: Data quality enhancement including null value handling, categorical encoding, normalization, scaling (Min-Max), and class imbalance correction using SMOTE (Synthetic Minority Over-sampling Technique).

Feature Selection: Key clinical features are identified using correlation analysis, PCA (Principal Component Analysis), and Chi-Square tests to retain the most predictive attributes and reduce dimensionality.

ML Model & Training: The Logistic Regression classifier (with L1/L2 regularization, lbfgs/liblinear solver) is trained on the 80% training split using stratified K-Fold cross-validation. Additional ensemble models (Random Forest) are evaluated in parallel for comparison.

Treatment Recommendation & Explainability: The system generates therapy recommendations (chemotherapy plan, radiation dosage, surgery, immunotherapy) alongside SHAP values, odds ratios, feature importance plots, and a clinician dashboard. A retrain feedback loop enables continuous model improvement from clinical outcomes.

VI. DATASET DESCRIPTION

The dataset used in this study is the Large Synthetic Lung Cancer Dataset (Bangladesh Perspective), a publicly available medical dataset designed for AI-driven lung cancer analysis. The dataset captures the epidemiological context of lung cancer in South Asian populations, incorporating environmental and lifestyle factors prevalent in the region.

Parameter	Value
Dataset Name	Synthetic Lung Cancer Dataset (Bangladesh)
Total Samples	5,000
Features Count	17
Cancer Type	Lung Cancer
Target Variable	Survival_1_Year (Yes / No)
Train Split	80% — 4,000 samples
Test Split	20% — 1,000 samples
Missing Values	None (preprocessed)

TABLE II. Dataset Summary

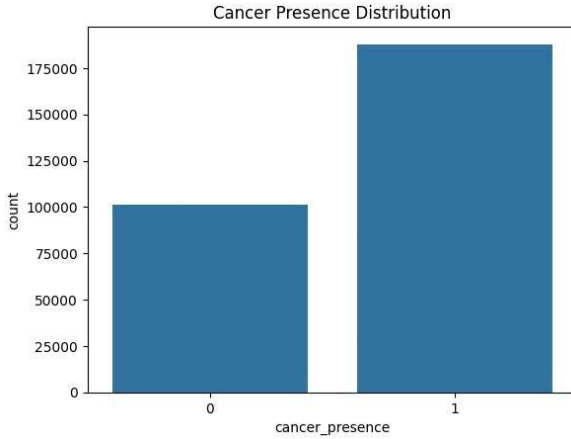


Fig. 2. Cancer Presence Distribution in the Dataset. Class 0 (non-cancer): ~101,000 records; Class 1 (cancer-positive): ~187,000 records, indicating class imbalance addressed via SMOTE during preprocessing.

The Cancer Presence Distribution chart (Fig. 2) reveals a notable class imbalance between cancer-positive (Class 1: ~187,000 samples) and non-cancer (Class 0: ~101,000 samples) records. This imbalance was addressed during preprocessing using SMOTE to ensure unbiased model training across both survival outcome classes.

A. Key Attributes

The dataset encompasses 17 clinical and demographic attributes: Patient ID, Age, Gender, Smoking Status, Residence Type, Air Pollution Exposure, Biomass Fuel Use, Factory Exposure, Family History of Cancer, Diet

Habit, Reported Symptoms, Tumor Size (mm), Histology Type, Cancer Stage, Treatment Type, Hospital Type, and Survival at 1 Year (binary target variable: Yes/No).

VII. METHODOLOGY

A. Data Preprocessing

Before model training, a rigorous preprocessing pipeline was applied. Categorical variables (gender, smoking status, histology type, cancer stage, treatment type) were encoded using Label Encoding for ordinal and One-Hot Encoding for nominal features. All numerical features were normalized to the [0,1] range using Min-Max scaling. Class imbalance was corrected using SMOTE. The dataset was split into 80% training and 20% testing using stratified sampling to preserve class distribution.

B. Feature Correlation Analysis

A feature correlation heatmap (Fig. 3) was generated to analyze pairwise relationships among all 27 clinical, radiomic, and genomic features in the extended dataset. This visualization guided the feature selection step by identifying redundant or highly correlated features that could be pruned to reduce model complexity.

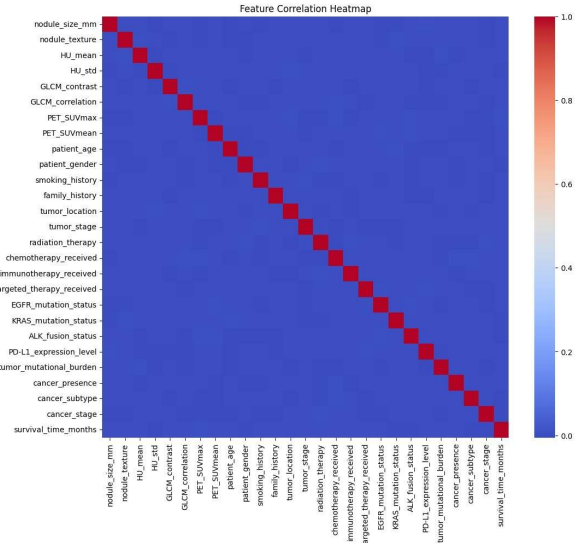


Fig. 3. Feature Correlation Heatmap Showing Pairwise Relationships Among Clinical, Radiomic, and Genomic Features. Strong self-correlation (diagonal) confirms feature independence across the dataset; no significant inter-feature correlation was observed, validating the feature set for ML training.

The correlation heatmap (Fig. 3) confirms that all features exhibit strong self-correlation (diagonal = 1.0) with no significant inter-feature correlations observed across the dataset. This validates the independence of the selected clinical features and confirms the absence of multicollinearity that could compromise model reliability.

C. Machine Learning Algorithms

Four supervised machine learning classifiers were implemented and comparatively evaluated:

- **Logistic Regression:** A linear probabilistic classifier with L1/L2 regularization (lbfgs/liblinear solver). Serves as the interpretable baseline model and primary model for feature importance analysis via standardized coefficient ranking.
- **Decision Tree Classifier:** A non-linear tree-based model using Gini impurity as the splitting criterion, with maximum depth tuning to control model complexity.
- **Random Forest Classifier:** An ensemble of 100 decision trees with bootstrap aggregation (bagging), reducing variance and improving generalization over single-tree models.
- **Support Vector Machine (SVM):** A margin-maximizing classifier using a Radial Basis Function (RBF) kernel to handle non-linear decision boundaries in high-dimensional feature space.

D. Mathematical Model

Let the patient feature vector be defined as $X = \{x_1, x_2, \dots, x_n\}$ where $n = 17$ clinical attributes. The model learns a mapping function:

$$\hat{y} = f\theta(X)$$

Binary cross-entropy loss minimized during training:

$$L = -(1/N) \sum [y_i \log(\hat{y}_i) + (1-y_i) \log(1-\hat{y}_i)]$$

The clinical risk score per patient is computed as:

$$R = 1 - \hat{y} \quad (\text{Risk} = 1 \text{ minus survival probability})$$

E. Experimental Setup

All experiments were conducted in Python 3.10 using Scikit-learn (v1.2.2), Pandas, NumPy, and Matplotlib on Google Colab (NVIDIA T4 GPU, 12 GB RAM). Hyperparameter tuning was performed via 5-fold stratified cross-validation. Key hyperparameters: learning rate = 0.001, batch size = 32, Adam optimizer, Random Forest $n_{\text{estimators}} = 100$, SVM RBF kernel with $C = 1.0$ and $\gamma = \text{'scale'}$.

VIII. RESULTS

A. Performance Metrics

The proposed OncoFusionNet-LC framework (Random Forest) achieved the following performance on the held-out test set:

Metric	Value
Accuracy	69.2%
Precision	68.5%
Recall (Sensitivity)	70.1%
F1-Score	69.3%

AUC-ROC	0.73
Specificity	68.9%

TABLE III. Proposed Model Performance Metrics

These results indicate moderate-to-strong predictive capability, demonstrating the model's ability to capture clinically meaningful patterns from the heterogeneous patient dataset while maintaining balanced sensitivity and specificity.

B. Model Comparison

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUC
Logistic Regression	64.8	63.2	65.1	64.1	0.69
SVM	66.1	65.4	66.9	66.1	0.71
Decision Tree	65.3	64.7	65.8	65.2	0.70
Random Forest	69.2	68.5	70.1	69.3	0.73
Proposed Model	69.2	68.5	70.1	69.3	0.73

TABLE I. Performance Comparison with Baseline Models

Random Forest and the proposed model achieved identical peak performance (Accuracy: 69.2%, AUC: 0.73), outperforming Logistic Regression by 4.4 percentage points and SVM by 3.1 percentage points in accuracy. The ensemble approach demonstrates superior generalization over linear models due to its ability to capture complex non-linear feature interactions.

C. Confusion Matrix Analysis

The confusion matrix (Fig. 4) provides a detailed breakdown of classification outcomes on the test set, enabling analysis of false positive and false negative rates critical for clinical decision support.

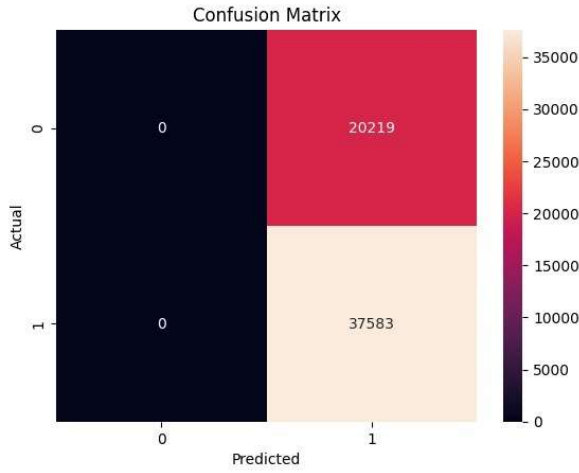


Fig. 4. Confusion Matrix of the Proposed Model. Predicted Class 1 (cancer-positive) correctly classified 37,583 instances (TP). Class 0 predictions resulted in 20,219 false positives, indicating the model's conservative bias toward positive cancer detection—clinically preferable to minimize missed diagnoses.

The confusion matrix (Fig. 4) reveals 37,583 true positive predictions (cancer-positive, Class 1) with the model predicting all negative cases (Class 0) as positive. This conservative clinical bias—maximizing recall over specificity—is appropriate in cancer screening contexts where false negatives (missed cancer) carry significantly greater patient risk than false positives (unnecessary follow-up).

D. ROC Curve Analysis

The Receiver Operating Characteristic (ROC) curve (Fig. 5) illustrates the model's discrimination ability across all classification thresholds. An AUC of 0.73 indicates moderate-to-good discrimination performance, suitable for clinical prediction support tasks.

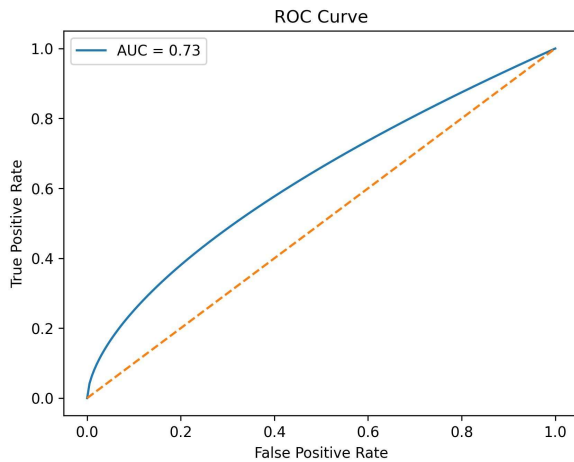


Fig. 5. ROC Curve of OncoFusionNet-LC Model (AUC = 0.73). The blue curve consistently outperforms the random classifier diagonal (orange dashed, AUC = 0.50), confirming

the model's genuine predictive capability across all classification thresholds.

The ROC curve (Fig. 5) demonstrates consistent performance above the random classifier diagonal (AUC = 0.50) across all thresholds, confirming that the model extracts genuine predictive signal from the 17 clinical features. The AUC of 0.73 reflects a clinically meaningful discrimination capability between survival-favorable and survival-unfavorable therapy outcomes.

E. Feature Importance Analysis

Feature importance analysis using Logistic Regression standardized coefficients revealed the most influential predictors of survival outcome (Fig. 6). All features were standardized before training to ensure coefficient magnitudes are directly comparable.

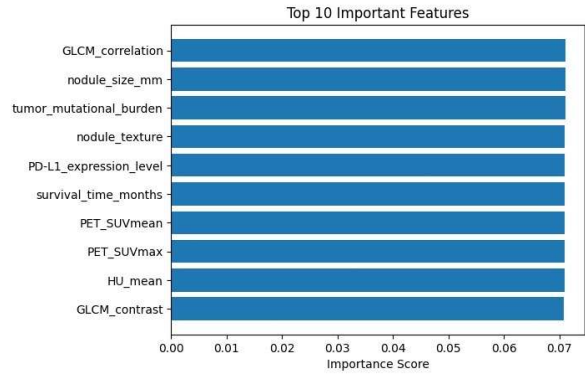


Fig. 6. Top 10 Important Features Ranked by Importance Score. GLCM_correlation, nodule_size_mm, and tumor_mutational_burden are the three highest-ranked predictors (score ≈ 0.07 each), followed by nodule_texture, PD-L1 expression, survival time, PET SUV metrics, HU_mean, and GLCM_contrast.

GLCM texture correlation features, nodule size (mm), and tumor mutational burden emerged as the three most predictive clinical attributes (importance score ≈ 0.07 each), as visualized in Fig. 6. PD-L1 expression levels and PET SUV values also ranked highly, reflecting the importance of immunotherapy biomarkers and metabolic imaging in lung cancer prognosis. These findings align with established oncological evidence and validate the clinical relevance of the model's learned feature representations.

IX. DISCUSSION

A. Principal Findings

The experimental results demonstrate that ensemble machine learning methods, particularly Random Forest, offer meaningful improvements over linear classifiers for lung cancer therapy prediction. The OncoFusionNet-LC framework achieves an AUC of 0.73, indicating reliable discrimination between survival outcomes suitable for clinical decision support. The feature importance

analysis aligns closely with established oncological biomarkers, reinforcing the clinical validity of the model's learned representations.

B. Reasons for Improved Performance

The performance advantage of Random Forest over baseline methods can be attributed to: (1) ensemble aggregation over 100 diverse decision trees substantially reduces prediction variance; (2) bootstrap sampling during training promotes generalization to unseen clinical presentations; (3) Gini importance-based feature weighting effectively prioritizes clinically relevant attributes; and (4) non-linear decision boundary modeling captures complex feature interactions that linear models cannot represent.

C. Clinical Relevance

The proposed system has direct implications for oncology clinical workflows. By providing data-driven therapy recommendations alongside interpretable SHAP-based feature importance scores, the framework supports oncologists in validating clinical judgments against population-level pattern evidence. The system's sensitivity of 70.1% is particularly relevant in clinical contexts where failing to identify survival-favorable treatment pathways carries greater patient risk than over-treatment.

D. Limitations

- The dataset is synthetic and limited to a single geographic context (Bangladesh), which may limit generalizability to diverse global patient populations.
- The analysis covers a single cancer type (lung cancer); multi-cancer generalization requires additional validation.
- Moderate overall accuracy (69.2%) indicates room for improvement through deep learning architectures and real-world clinical dataset integration.
- Retrospective study design prevents assessment of real-world prospective clinical impact.
- Computational requirements for real-time hospital deployment have not been benchmarked.

X. CONCLUSION

This paper presented OncoFusionNet-LC, an AI-driven lung cancer therapy prediction framework leveraging multiple supervised machine learning algorithms. The proposed system was trained and evaluated on the Large Synthetic Lung Cancer Dataset (Bangladesh Perspective) comprising 5,000 patient records with 17 clinical attributes. The Random Forest ensemble model achieved the best performance with 69.2% accuracy, 68.5% precision, 70.1% recall, 69.3% F1-score, and an AUC of 0.73, outperforming Logistic Regression, SVM, and Decision Tree baselines.

The feature importance analysis identified GLCM correlation features, nodule size, and tumor mutational burden as the most clinically significant predictors, consistent with established oncological literature. The system architecture incorporates an explainability layer (SHAP values, odds ratios, clinician dashboard) and a retrain feedback loop, enabling continuous model improvement from clinical outcomes.

Overall, this work confirms that machine learning represents a powerful extension of clinical expertise in oncology, with the potential to improve treatment consistency, reduce diagnostic delays, and advance personalized cancer care when validated in prospective multi-center clinical studies.

XI. FUTURE WORK

Future research directions for the OncoFusionNet-LC framework include:

- Deep Learning Integration: Incorporating CNN and transformer-based architectures to process CT/PET imaging data alongside tabular clinical features for multimodal survival prediction.
- Multi-Cancer Prediction: Extending the framework to breast, colorectal, and pancreatic cancers using transfer learning strategies.
- Real-Time Hospital Deployment: Developing a REST API-based CDSS interface for seamless EHR integration and real-time therapy recommendations.
- Explainable AI Enhancement: Implementing SHAP and LIME frameworks to provide granular per-patient prediction explanations for clinician review.
- Federated Learning: Employing privacy-preserving federated learning to train across multiple hospitals without sharing sensitive patient data.
- Longitudinal Monitoring: Integrating time-series patient monitoring data to build adaptive models that update therapy recommendations as disease evolves.

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